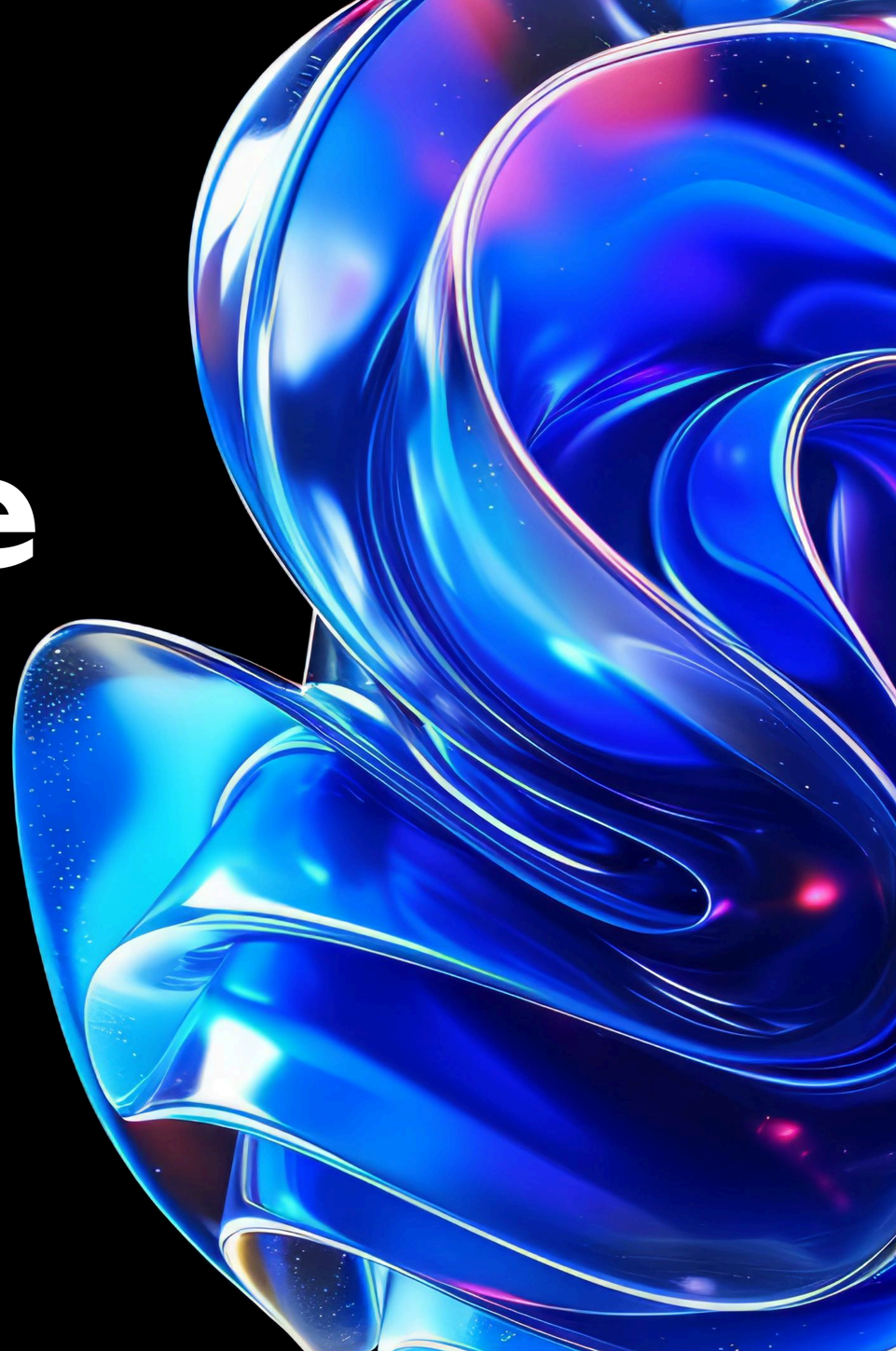
ML for Cognitive Performances

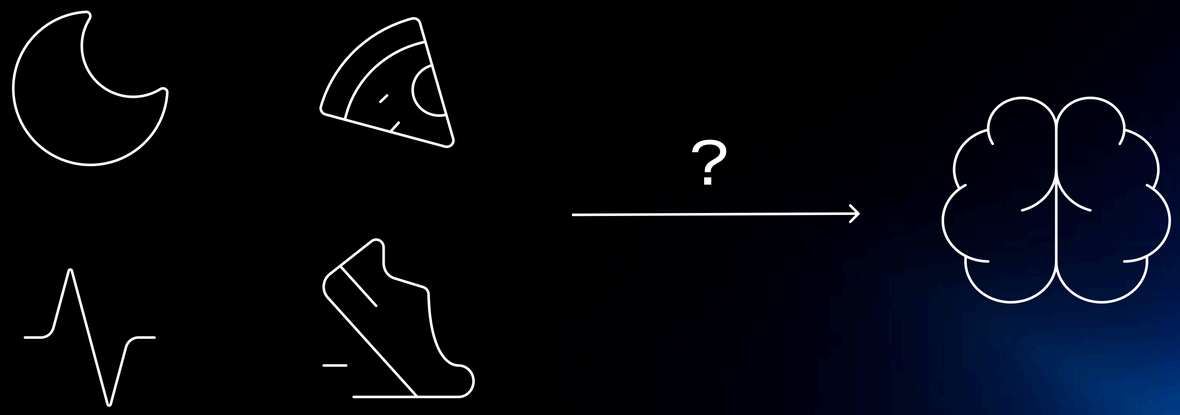
Cognitive and Athletic Performance Optimization using Machine Learning and Biometrics



Edoardo Sorgentone & Luis Mariduena

01 — The Problem

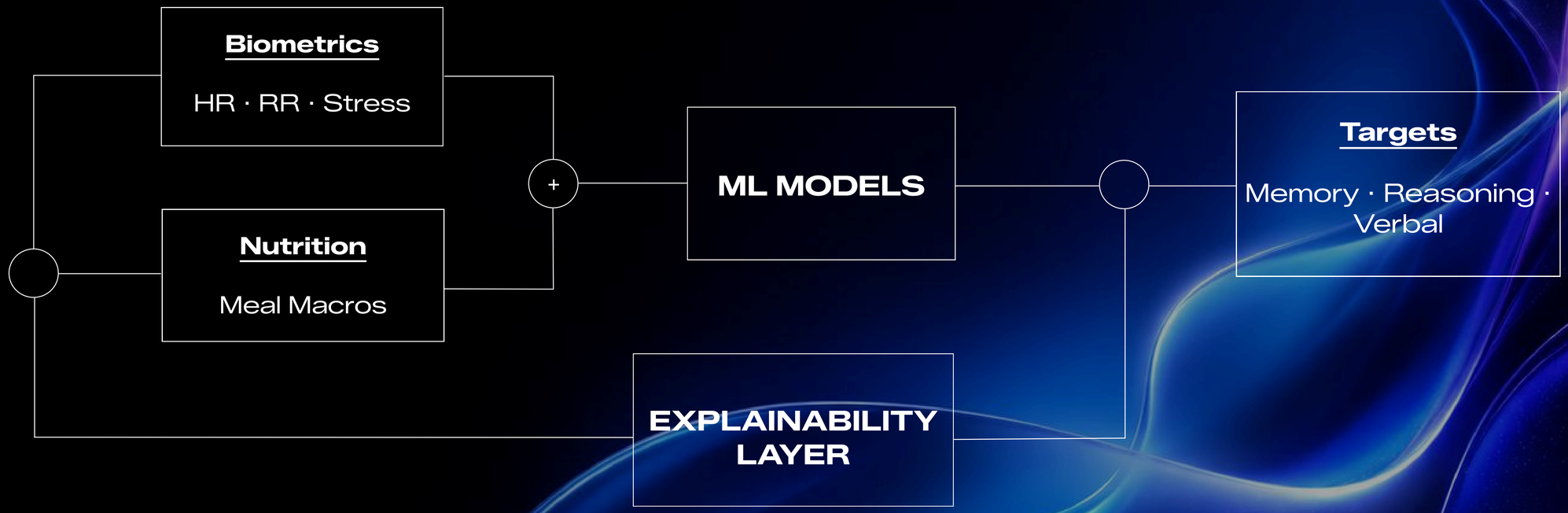
Cognitive performance is influenced by multiple physiological and behavioral factors: sleep, nutrition, heart rate, respiration and stress.



Can daily biometric and lifestyle signals explain cognitive performance?

02 — The Solution

A two-stage pipeline for cognitive performance prediction and model explainability.



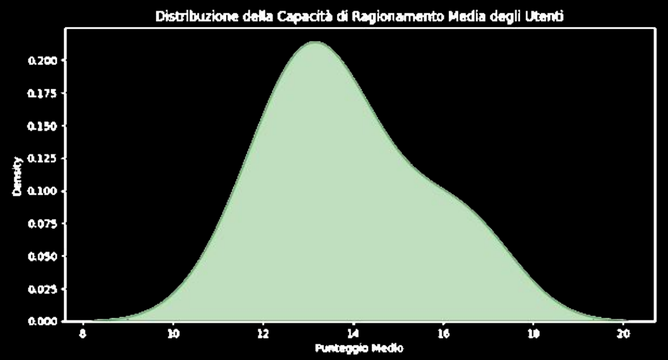
The system first selects the best model configuration for each cognitive target, then applies SHAP, ALE and DiCE to interpret feature effects and generate actionable insights.

03 — ML Pipeline

1. Pre-processing & EDA

Global and subject-level EDA on cognitive targets and biometric variables.Inter-subject variability highlighted different cognitive baselines and physiological patterns.

Dataset merge(nutritional + biometrics), categorical encoding, target filtering and subject-aware missing-value handling.Quantile-based discretization produced balanced target classes, so no class weighting was required.

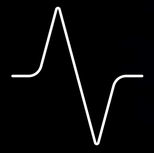


2. Feature engineering

Raw HR and RR time-series from wearable biometric dataset were converted into fixed-length descriptors.Extracted statistics: mean, std, min/max, skewness, kurtosis, IQR, RMS and energy.

Temporal and peak-based features captured short-term physiological variability.

Subject-wise robust scaling was applied to dynamic features using median and inter-quartile range, reducing inter-subject baseline effects and avoiding data leakage.



→ [Mean,Std,Min,Max,RMS, ...]

3. LOSO Cross Validation

One subject is held out at each fold and used as test set.

This prevents subject-level leakage and evaluates generalization to unseen users.

LOSO

```
scores = []

for subject in subjects:

    D_test  = samples_of(subject)
    D_train = samples_not_of(subject)

    train(model, D_train)

    score = evaluate(model, D_test)

    scores.append(score)

return mean(scores)
```

04 —ML Pipeline

4. Feature Selection

- **Forward selection** starts from an empty feature set and adds the most useful feature at each step.
- **Backward selection** starts from all features and removes those that reduce validation quality.
- **Stepwise selection** combines forward addition and backward removal.

5. Hyperparameter Optimization

For each model–feature subset pair, Optuna optimized model-specific hyperparameters using LOSO validation accuracy as objective.

6. Model Selection Algorithm

We defined a model array containing all candidate models.For every target and FS method, the algorithm loops through the model array:

1. initialize the model
2. select the best feature subset
3. run Optuna on that subset
4. evaluate with LOSO accuracy
5. store features, best params and score.

Best Configurations:

Target	Best Model	Accuracy
Memory	LightGBM	0.639
Reasoning	SVC	0.718
Verbal	SVC	0.664

PSEUDO-CODE MODEL SELECTION ALGO

```
for target in cognitive_targets:
    y = select_target(target)

    for fs_method in FS_methods:
        for model in candidate_models:

            features = select_features(
                X, y, model, fs_method
            )

            best_params, score = optuna_optimize(
                model=model,
                X=X[features],
                y=y,
                validation="LOSO"
            )

            store_result(
                target, model, fs_method,
                features, best_params, score
            )

return best_model_per_target(results)
```

05 — Explainability Pipeline

1. Data Reception

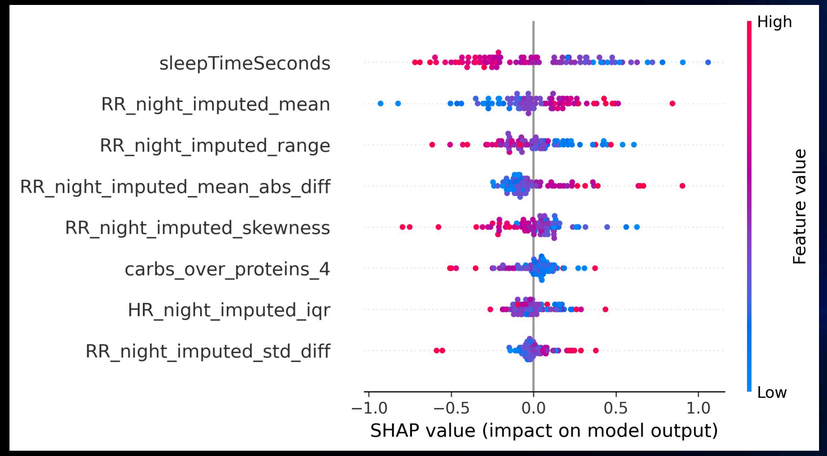
Receives the 2 best configurations for each target area. Including their features,best params,score and the associated final data sets.

2. SHAP Analysis

Utilizes received data on the model configurations to:

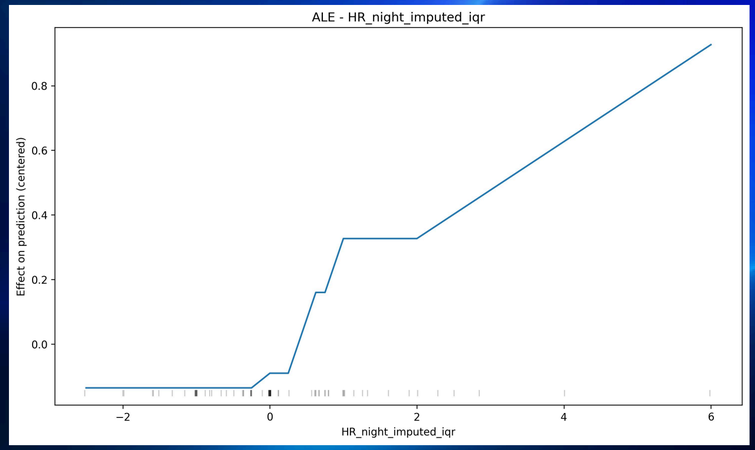
1. Calculate SHAP Values
2. Generate Beeswarm Plots
3. Generate Bar plots
4. Generate Water Fall plots
5. Store the graphs in a google drive folder for later analysis
6. Select which features will be analyzed by the next two tools

Repeats the process for all the selected Models. With the goal of showing the local effects of features, and their importance.



3. ALE Analysis

Calculates accumulated local effects of the 7 best features of each given configurations as given by their mean(|SHAP Values|) then generates and ALE plot for each with the goal of showing global effects of the features.



06 — Explainability Pipeline

1. **DiCe Counterfactual Generation**

Selects the 5 best features of each given configuration base on their mean(|SHAP Values|) then:

1. Selects 9 random results of the original dataset including feature values
2. Generates 5 counter-factuals that would change the score to one of the other 2 possibilities according to the given configuration
3. Store all the resulting data ordered as follows: Per goal target score row 1 original result, next 5 rows the counter-factuals generated with the goal of predicting a different target result.
4. Stores the results into a drive folder on a per configuration and target basis

Repeats for all the selected configurations

Format better visible on the table on the right

Feature Values	Type	Target	Modified Features
OG Values	Original	1	Nan
CF Val 1	CF	2	Mod list 1
CF Val 2	CF	2	Mod list 2
CF Val 3	CF	2	Mod list 3
CF Val 4	CF	2	Mod list 4
CF Val5	CF	2	Mod list 5

07 — Explainability Pipeline

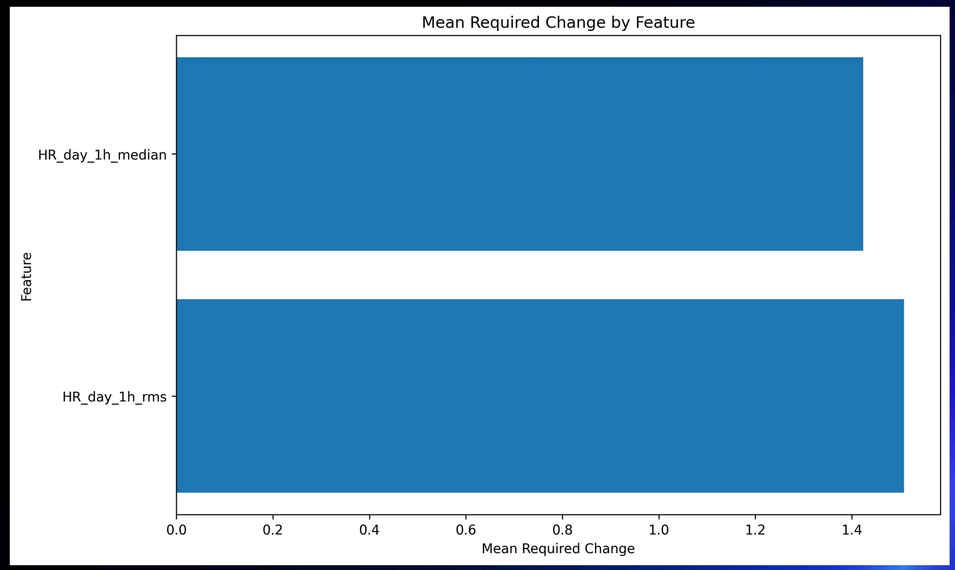
1. Counterfactual Data Processing

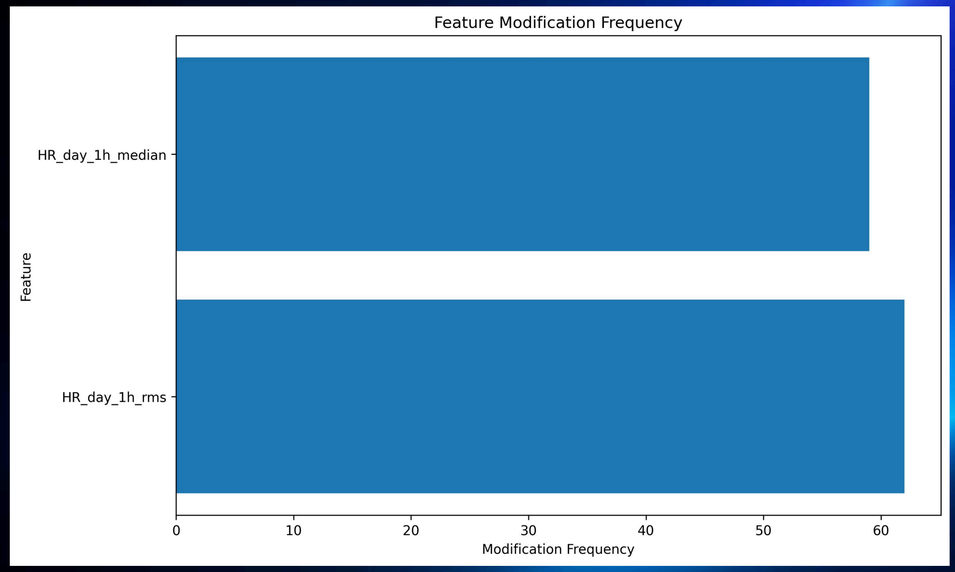
Converts hard to interpret .csv table data into two types human interpretable graphs

1. A feature modification frequency graph to see whether DiCe results agree with SHAP analysis on which are the most important features
2. A Mean Required feature change graph to see how big the modification of the features had to be to achieve given goals.

The goal of this generated data is to corroborate the findings of the two more easily interpretable analyses. Rather than permitting us to make new ones.

Thise because we had the limitation of our best SVC configurations being trained without a probability function which is necessary for DiCe Generation.





08 —Conclusions

1. Counterfactual Data Processing

After extensive analysis of all the data generated by the Explainability pipeline we were able to arrive to a couple conclusions valid for all the cognitive targets.

1. There is a strong positive correlation between sleep quality and quantity, and cognitive performance with them being recurring important features in all configurations

2. Higher daily exercise also shows a possible positive correlation with cognitive performance, as fig.1 shows Day HR variability: there's a positive trend.

3. And notably possible negative correlations between high fat on meal 5(fig.2 shows fat intake: negative trend) and, verbal and reasoning performance. As well possible negative impact of a higher carbohydrate to protein ratio on verbal performance. Though these can't be confirmed as surely and would need a larger data pool to corroborate

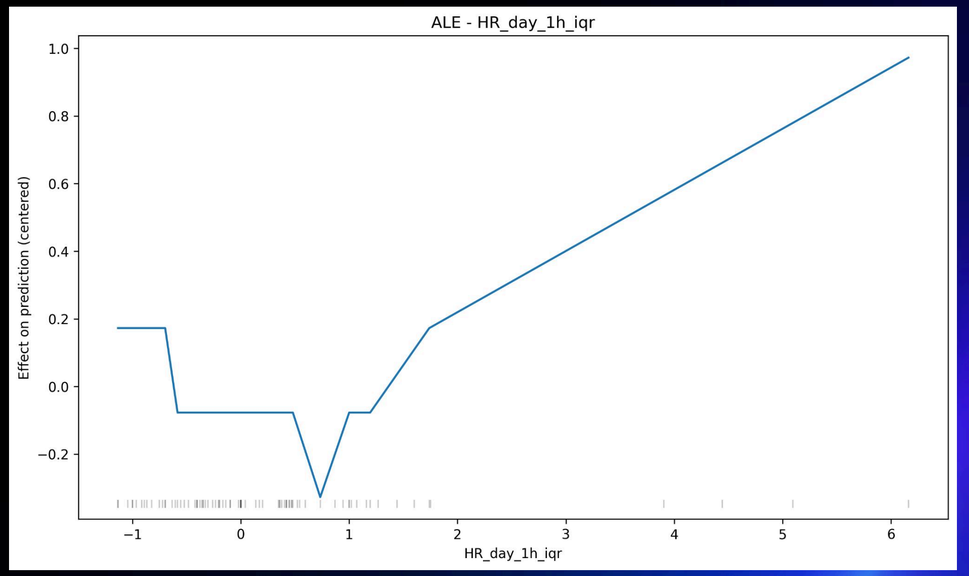


fig.1

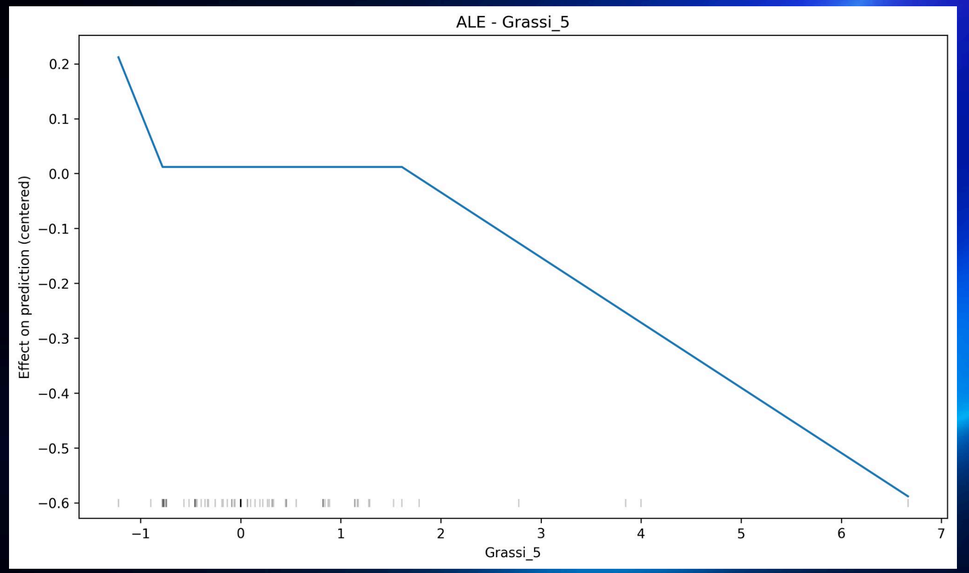


fig.2

Thank you